SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Computer Vision with OpenCV COURSE CODE: DSA0202

COURSE OUTCOME COVERED
CO2: *Apply image processing and feature extraction techniques, including edge detection, motion estimation, and optical flow, for analyzing images.. (BL3)*

Assessment Tool Weightage: 20%

QUESTIONS: 2 Total Marks:20 Annexure A

Class Test

Code of Conduct:

I M Tilak Venkat Sai(192424415)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**	**Marks**
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions	
Explanation	25%	Detailed, well explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation	
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis	
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers	
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps	

Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand	

Annexure A

Class Test

Question 1: Image Preprocessing and Feature Extraction

Scenario: A drone is used to capture aerial images of a farmland to identify pest affected crop regions. Due to sensor limitations, wind movement, and changing sunlight, the captured images contain noise, shadows, and uneven illumination. Pest affected areas usually show visible texture and color variations compared to healthy crop regions.

Answer the following questions:

a) Preprocessing Pipeline — 4 Marks

Propose a suitable preprocessing pipeline to improve the quality of the farmland images before analysis. Your answer should include methods for noise reduction, illumination correction, contrast enhancement, and normalization. Justify the purpose of each step.

b)Feature Extraction — 3 Marks

Choose an appropriate feature extraction method to identify texture changes caused by pest damage. Explain why the selected method is suitable for detecting affected crop regions.

c)Rotation and Scale Variations — 3 Marks

Drone images may be captured from different heights and angles. Discuss how rotation and scale variations can affect feature detection. Suggest suitable methods to make the feature extraction process more robust.

Question 2: Stereo Vision and Motion Estimation

Scenario: A self-driving car is equipped with stereo cameras to navigate through a busy street. The system must estimate the distance of nearby vehicles and pedestrians in real time. It should also track moving objects accurately to avoid collisions, especially when pedestrians or vehicles change speed suddenly.

Answer the following questions:

a) Stereo Disparity and Depth Estimation — 4 Marks

Explain how stereo cameras use the difference between left and right image views to estimate the depth of objects. Discuss how disparity changes for nearby and distant objects.

b) Optical Flow for Motion Tracking — 3 Marks

If a pedestrian suddenly changes speed or direction, explain how optical flow helps in tracking the pedestrian's motion between consecutive video frames. c) Errors and Mitigation Techniques — 3 Marks

Identify possible sources of error in depth estimation and motion tracking, such as camera misalignment, poor lighting, occlusion, fast motion, or textureless regions. Suggest suitable techniques to reduce these errors.

Question 1: Image Preprocessing and Feature Extraction

a) Preprocessing Pipeline

Preprocessing Steps:

1. **Image Acquisition** – Capture aerial farmland images using a drone.
2. **Noise Reduction** – Apply a Gaussian Filter or Bilateral Filter to remove sensor noise while preserving edges.
3. **Illumination Correction** – Use CLAHE (Contrast Limited Adaptive Histogram Equalization) or Histogram Equalization to reduce shadows and uneven lighting.
4. **Contrast Enhancement** – Apply Contrast Stretching or CLAHE to improve the visibility of pest-affected regions.
5. **Normalization** – Normalize pixel intensity values to maintain consistent brightness across all images.
6. **Color Space Conversion (Optional)** – Convert RGB to HSV or Lab color space to better distinguish healthy and affected crops.

Justification:

- Noise reduction removes unwanted sensor noise.
- Illumination correction compensates for changing sunlight and shadows.
- Contrast enhancement highlights pest-damaged regions.
- Normalization improves consistency between images.
- HSV/Lab color spaces separate color information more effectively.

b) Feature Extraction

Suitable Method: Local Binary Pattern (LBP)

Reason:

- Captures texture variations effectively.
- Detects differences between healthy and pest-affected crop regions.

- Robust to changes in illumination.
- Computationally efficient for real-time analysis.

(SIFT can also be mentioned if feature matching is required.)

c) Rotation and Scale Variations (3 Marks)

Effect:

- Different drone heights change image scale.
- Different camera angles rotate image features.
- These variations reduce feature matching accuracy.

Robust Methods:

- **SIFT** – Scale and rotation invariant.
- **SURF** – Faster than SIFT with similar robustness.
- **ORB** – Rotation invariant and suitable for real-time applications.

Question 2: Stereo Vision and Motion Estimation

a) Stereo Disparity and Depth Estimation

Stereo cameras capture the same scene from two slightly different viewpoints.

Working:

1. Capture left and right images.
2. Match corresponding points in both images.
3. Compute the disparity (difference in pixel position).
4. Estimate depth using disparity.

Relationship:

- **Large disparity → Object is closer.**
- **Small disparity → Object is farther away.**

Thus, depth is **inversely proportional to disparity**.

b) Optical Flow for Motion Tracking

Optical flow estimates the movement of pixels between consecutive video frames.

If a pedestrian suddenly changes speed or direction:

- Optical flow vectors change in magnitude and direction.
- The system updates the pedestrian's new position.
- Continuous frame-to-frame tracking helps maintain accurate motion estimation and

collision avoidance.

c) Errors and Mitigation Techniques (3 Marks)

Source of Error	**Mitigation Technique**
Camera Misalignment	Stereo camera calibration and image rectification
Poor Lighting	CLAHE, Histogram Equalization, Image Enhancement
Occlusion	Kalman Filter or Particle Filter for motion prediction
Fast Motion	High frame-rate cameras and Pyramid Lucas-Kanade Optical Flow
Textureless Regions	Feature detectors such as SIFT, SURF, or ORB to improve matching

These techniques improve both **depth estimation** and **motion tracking accuracy** in self-driving applications.